for some $a_0, \ldots, a_{n-1} \in A$ and $s_0, \ldots, s_{n-1} \in S := A - \mathfrak{p}$. Multiplying both sides by s^n, where $s = s_0 \cdots s_{n-1} \in S$, shows that sx is integral over A, hence an element of A, since A is integrally closed. But then $sx/s = x$ is an element of $A_\mathfrak{p}$, so $A_\mathfrak{p}$ is integrally closed as claimed.

Thus (ii) implies that every $A_\mathfrak{p}$ is an integrally closed noetherian local domain of dimension at most 1, and for $\mathfrak{p} \neq (0)$ we must have $\dim A_\mathfrak{p} = 1$. Thus for every nonzero prime ideal $\mathfrak{p}$, the ring $A_\mathfrak{p}$ is an integrally closed noetherian local domain of dimension 1, and therefore a DVR, by Theorem 1.16. $\square$

Definition 2.10. A noetherian domain satisfying either of the equivalent properties of Proposition 2.9 is called a *Dedekind domain*.

Corollary 2.11. *Every PID is a Dedekind domain. In particular, $\mathbb{Z}$ is a Dedekind domain, as is $k[x]$ for any field k.*

Remark 2.12. Every PID is both a UFD and a Dedekind domain. Not every UFD is a Dedekind domain (consider $k[x, y]$, for any field k), and not every Dedekind domain is a UFD (consider $\mathbb{Z}[\sqrt{-13}]$, in which $(1 + \sqrt{-13})(1 - \sqrt{-13}) = 2 \cdot 7 = 14$). However (as we shall see), every ring that is both a UFD and a Dedekind domain is a PID.

One of our first goals in this course is to prove that the ring of integers of number fields and coordinate rings of global function fields are Dedekind domains. More precisely, we will prove that if A is a Dedekind domain and L is a finite separable extension of its fraction field, then the integral closure of A in L is a Dedekind domain. This includes the two main cases of interest to us, in which either $A = \mathbb{Z}$ and L is a number field, or $A = \mathbb{F}_q[t]$ and L is a global function field. Recall from Lecture 1 that number fields and global function fields are the two types of *global fields* (as we will prove in later lectures).

2.5 Fractional ideals

Throughout this subsection, A is a noetherian domain (not necessarily a Dedekind domain) and K is its fraction field.

Definition 2.13. A *fractional ideal* of a noetherian domain A is a finitely generated A-submodule of its fraction field.

Fractional ideals generalize the notion of an ideal: when A is noetherian the ideals of A are precisely the finitely generated A-submodules of A, and when A is also a domain we can extend this notion to its fraction field. Every ideal of A is also a fractional ideal of A, but fractional ideals are typically not ideals because they need not be contained in A. Some authors use the term *integral ideal* to distinguish the fractional ideals that lie in A (and are thus ideals) but we will not use this terminology.

Lemma 2.14. *Let A be a noetherian domain with fraction field K, and let $I \subseteq K$ be an A-module. Then I is finitely generated if and only if $aI \subseteq A$ for some nonzero $a \in A$.*

Proof. For the forward implication, if $r_1/s_1, \ldots, r_n/s_n$ generate I as an A-module, then $aI \subseteq A$ for $a = s_1 \cdots s_n$. Conversely, if $aI \subseteq A$, then aI is an ideal, hence finitely generated (since A is noetherian), and if $a_1, \ldots, a_n$ generate aI then $a_1/a, \ldots, a_n/a$ generate I. $\square$